



# CV2090

## Probability & Statistics for Civil Engineers

(Jul – Nov 2026)

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L11  
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# Messages from the Last Class

## Message 1: Probability density function (PDF)

$$Pr(a < X < b) = \int_a^b f_X(x) dx \quad f_X \geq 0$$
$$\int_{-\infty}^{\infty} f_X(x) dx = 1 \quad \rightarrow \text{Area under the density curve}$$

## Message 2: Cumulative distribution function (CDF)

$$F_X(x) = \int_{-\infty}^x f_X(x) dx$$

$$F_X(-\infty) = 0$$

$$F_X(+\infty) = 1$$

$$x < y \Rightarrow F_X(x) \leq F_X(y)$$

## Message 3: Mean $\mu$ aka $E[X]$

- long-term average
- metric of centrality
- weighted average, with  $w_{x_i} \equiv$

$$\mu = \sum_i x_i p_X(x_i)$$

$$\mu = \int_{-\infty}^{\infty} x f_X(x) dx$$

# Variance

- Which one is larger?

Average of squares  
or  
Square of averages

$$\begin{array}{c} E[X^2] \\ \vee \\ \mu^2 \end{array}$$

A metric for dispersion/spread.

choices

\* Deviation:  $X - \mu \Rightarrow E[X - \mu] = 0$

\* Mean Absolute deviation:  $|X - \mu|$   
 $\rightarrow$  error accumulates linearly.

\* Square of variation  $(X - \mu)^2$

$$\sigma^2 = E[(X - \mu)^2] = E[X^2] - \mu^2$$

# Variance

Discrete rv  $E[(X-\mu)^2] = \sum_i (x_i - \mu)^2 p_X(x_i)$

$$= \sum_i (x_i^2 + \mu^2 - 2\mu x_i) p_X(x_i)$$
$$= \sum_i x_i^2 p_X(x_i) + \mu^2 \underbrace{\sum_i p_X(x_i)}_{\rightarrow 1} - 2\mu \underbrace{\sum_i x_i p_X(x_i)}_{\rightarrow \mu}$$

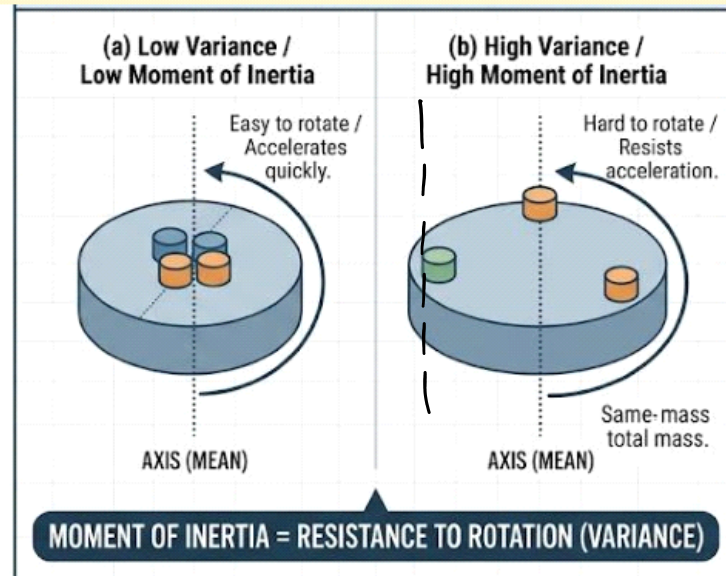
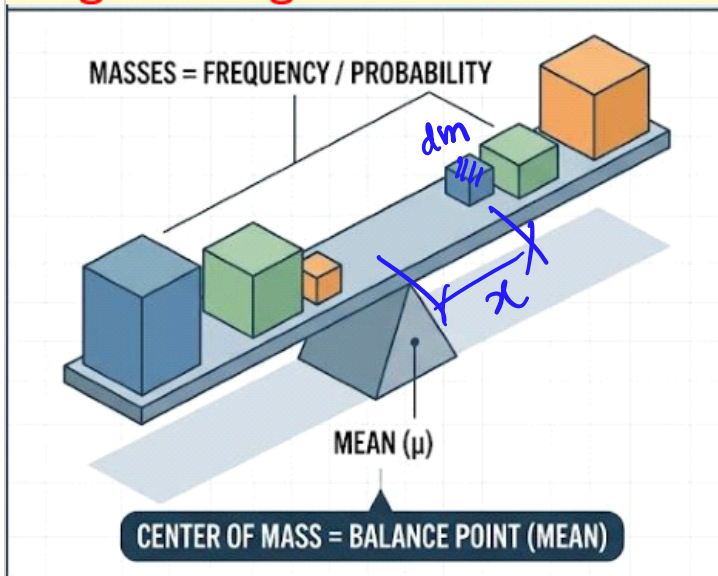
Cont. rv  $= E[X^2] - \mu^2$

→ @ home.

# Formula for Variance

$\{\text{Mean, Variance}\} \equiv \{\text{C.M., Moment of inertia}\}$

Engineering mechanics:



# Analogy with Engineering Mechanics

| Param.               | Mechanics                                                                                                       | Probability                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Mass                 | mass (or area)                                                                                                  | probability                                                                         |
| Value                | position, $x$                                                                                                   | RV $X$                                                                              |
| Density              | mass density, $\rho$                                                                                            | PDF, $f_X(x)$                                                                       |
| <b>First moment</b>  | $M_1 = \int x \, dm$ <p>center of mass = <math>\frac{M_1}{\int dm}</math></p> <p>where is the mass located?</p> | <p>first moment,</p> $\int_{-\infty}^{\infty} x f_X(x) \, dx = \mu$                 |
| <b>Second moment</b> | $M_2 = \int x^2 \, dm$ <p>moment of inertia</p> <p>how spread out is the mass?</p>                              | <p>second moment,</p> $\int_{-\infty}^{\infty} (x - \mu)^2 f_X(x) \, dx = \sigma^2$ |

# Mean, Variance of Different Distributions

| Distr.            | Description/Example                                                                                                                                                   | $p_X(x), \mu, \sigma^2$                                                                                                               |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Bernoulli         | $x$<br>A coin toss                                                                                                                                                    | $f_X(x) = p^x(1-p)^{1-x}, x = 0,1$<br>$\mu = p, \sigma^2 = pq$                                                                        |
| Binomial          | exactly $x$ success in $n$ trials<br>Prob. of 4 Heads on 10 coin flips?                                                                                               | $p_X(x) = \binom{n}{x} p^x q^{n-x}$<br>$\mu = np, \sigma^2 = npq$                                                                     |
| Geometric         | $x$ trials required for the first success<br>Prob. of getting first H on 10th toss?                                                                                   | $p_X(x) = (1-p)^{x-1} p$<br>$\mu = \frac{1}{p}, \sigma^2 = \frac{(1-p)}{p^2}$                                                         |
| Poisson           | $x$ events in a fixed time interval, when events occur independently at a constant average rate.<br>Prob. of 3 accidents at an intersection in a day?                 | $p_X(x) = \frac{e^{-\lambda} \lambda^x}{x!}, x = 0,1,2, \dots$<br>$\mu = \lambda, \sigma^2 = \lambda$                                 |
| Negative Binomial | $x$ trials needed to obtain $r$ successes<br>Prob. of getting 4 <sup>th</sup> Head on 10th coin toss?                                                                 | $p_X(x) = \binom{x-1}{r-1} (1-p)^{x-r} p^r$<br>$\mu = \frac{r}{p}, \sigma^2 = \frac{r(1-p)}{p^2}$                                     |
| Hypergeometric    | $x$ success in $n$ trials, from a population of size $N$ , which has $K$ successes (without replacement)<br>Prob. of getting two Aces in 4 picks without replacement? | $p_X(x) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}$<br>$\mu = np$ , where $p = K/N$<br>$\sigma^2 = np(1-p) \frac{N-n}{N-1}$ |

# Mean and Median Wealth

# Income disparity

Household Income (₹ lakh)

2 3 4 5 6 7 8 10 12 15 120

What is the typical household income?

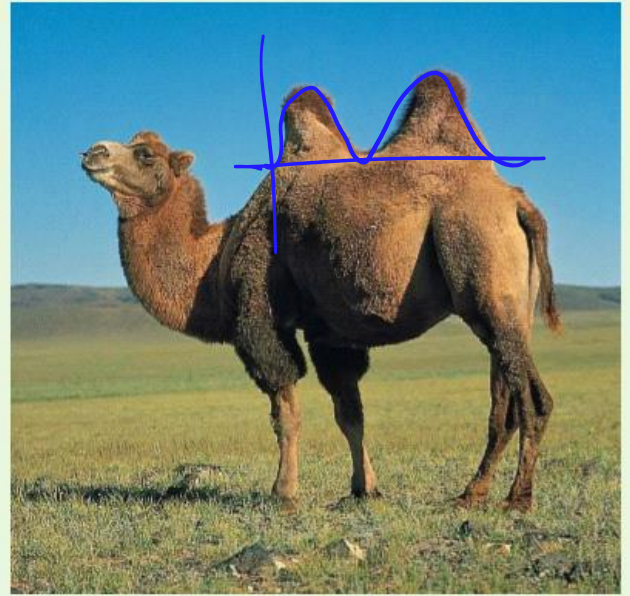
mean = 17.4 lakh

median = 7 lakh

# Mean versus Median



Arabic Camel



Mongolian Camel

# Higher Moments

# Expectation and Moments

Expected value of  $g(x)$

$$E[g(x)] =$$

①  $\int g(x) dx$

②  $\int g(x) f_x(x) dx$

③  $\int g(x) f_x(g(x)) dx$

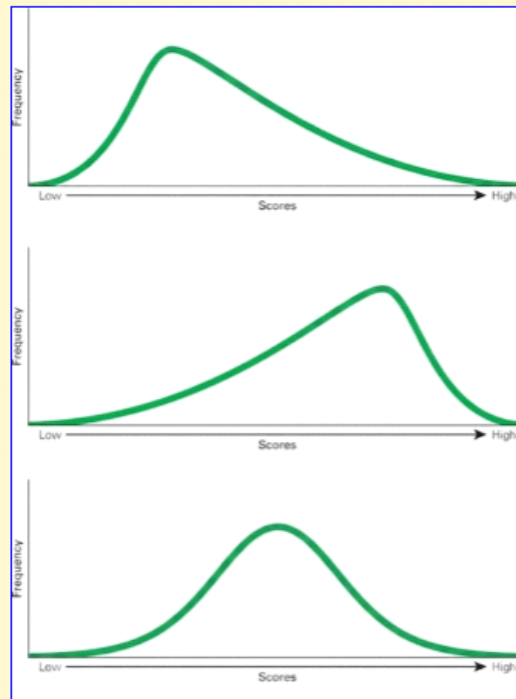
6000 time

|     |      |     |      |     |      |                  |
|-----|------|-----|------|-----|------|------------------|
| 500 | 1500 | 500 | 1500 | 500 | 1500 | $\leftarrow f$   |
| 1   | 2    | 3   | 4    | 5   | 6    | $\leftarrow x$   |
| 1   | 4    | 9   | 16   | 25  | 36   | $\leftarrow x^2$ |

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# Central Moments, Skewness

$\mu_x(3)$

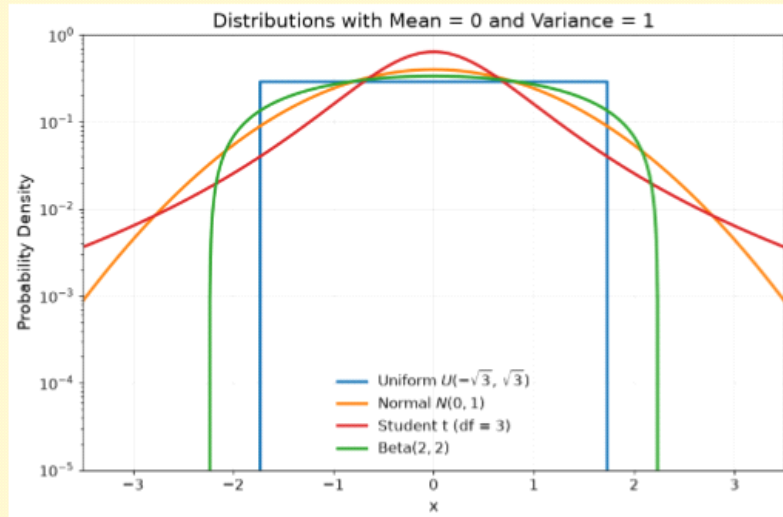
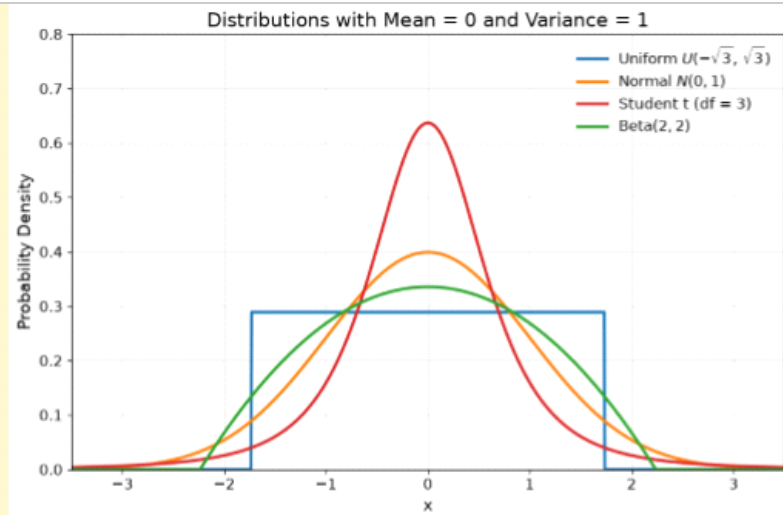


Qualitatively mark

- Mean,
- Median, and
- Mode,

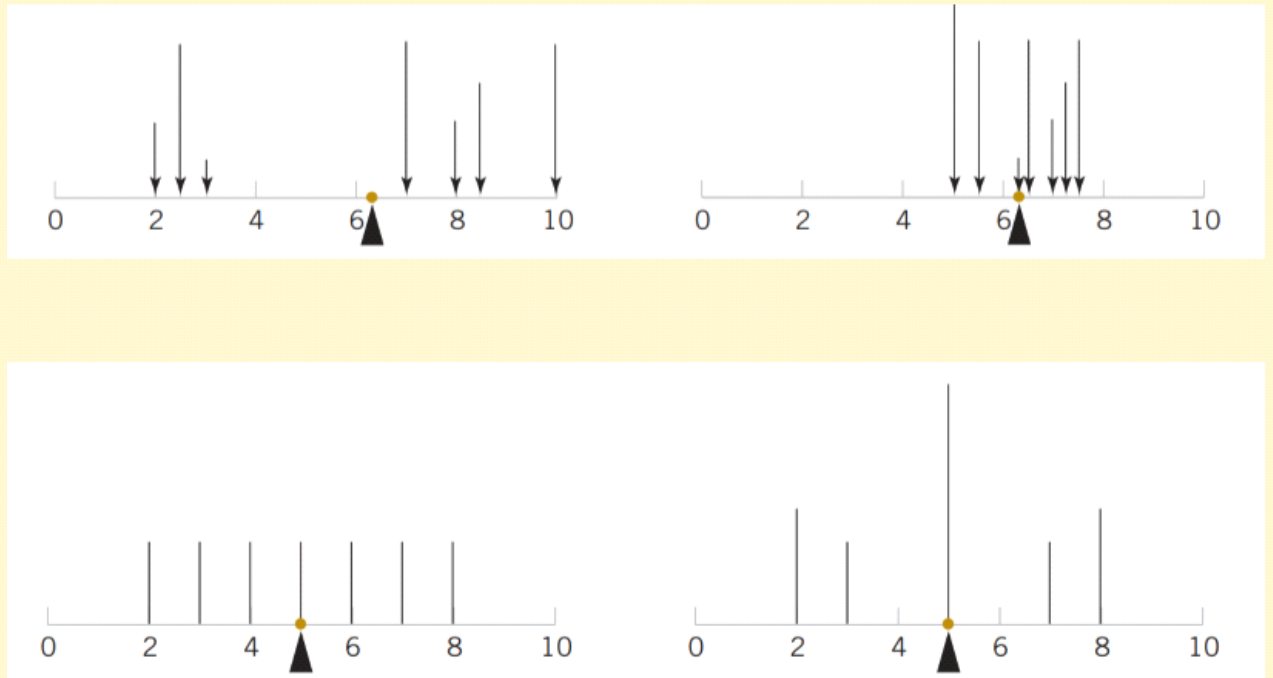
on all three figures.

# Kurtosis



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# {Mean, Variance} $\neq$ PMF/PDF



# Tyrannosaurus Rex (T-Rex)

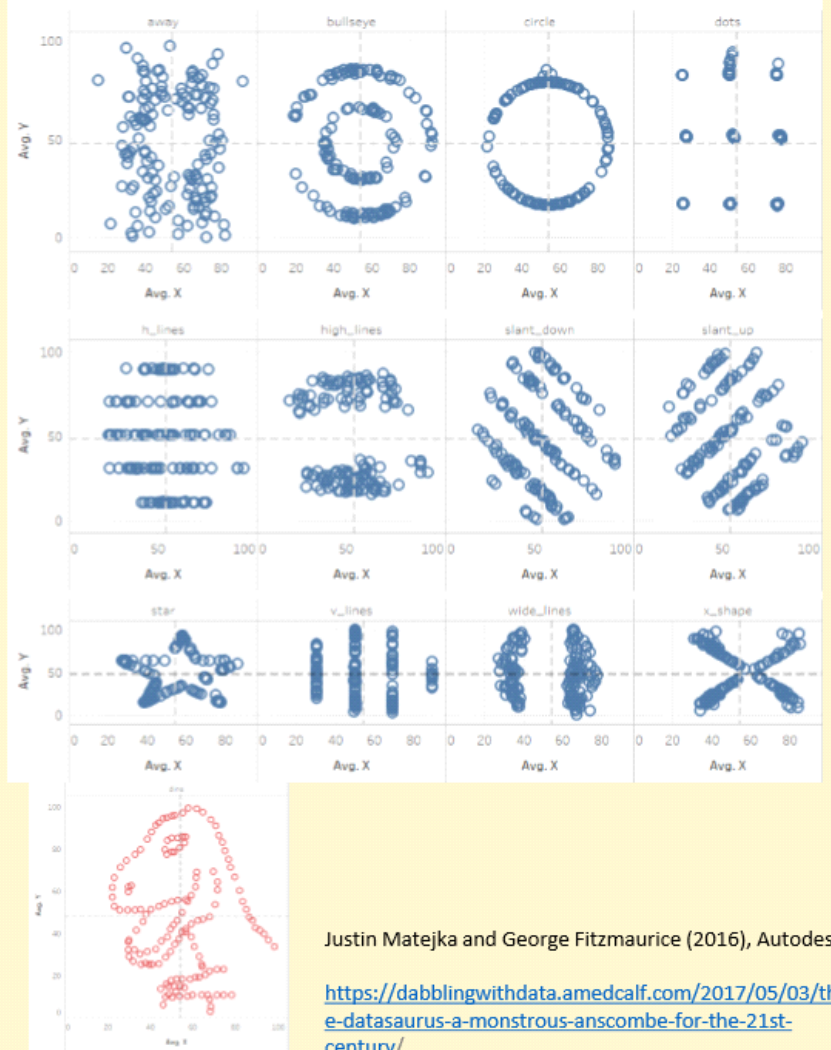


# Tyrannosaurus Rex (T-Rex)



# Datasaurus Dozen

Datasaurus Dozen

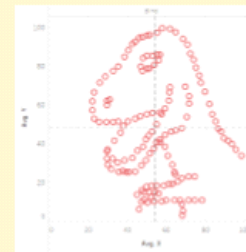
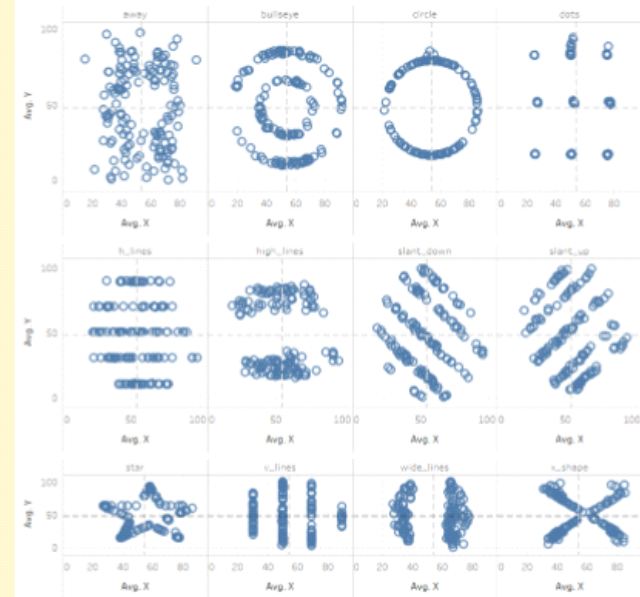


# Datasaurus Dozen

<https://seanfobbe.com/tutorials/importance-of-visualization-datasaurus/>

|     | dataset    | x_mean   | x_sd     | y_mean   | y_sd     | corr        |
|-----|------------|----------|----------|----------|----------|-------------|
|     | <char>     | <num>    | <num>    | <num>    | <num>    | <num>       |
| 1:  | dino       | 54.26327 | 16.76514 | 47.83225 | 26.93540 | -0.06447185 |
| 2:  | away       | 54.26610 | 16.76982 | 47.83472 | 26.93974 | -0.06412835 |
| 3:  | h_lines    | 54.26144 | 16.76590 | 47.83025 | 26.93988 | -0.06171484 |
| 4:  | v_lines    | 54.26993 | 16.76996 | 47.83699 | 26.93768 | -0.06944557 |
| 5:  | x_shape    | 54.26015 | 16.76996 | 47.83972 | 26.93000 | -0.06558334 |
| 6:  | star       | 54.26734 | 16.76896 | 47.83955 | 26.93027 | -0.06296110 |
| 7:  | high_lines | 54.26881 | 16.76670 | 47.83545 | 26.94000 | -0.06850422 |
| 8:  | dots       | 54.26030 | 16.76774 | 47.83983 | 26.93019 | -0.06034144 |
| 9:  | circle     | 54.26732 | 16.76001 | 47.83772 | 26.93004 | -0.06834336 |
| 10: | bullseye   | 54.26873 | 16.76924 | 47.83082 | 26.93573 | -0.06858639 |
| 11: | slant_up   | 54.26588 | 16.76885 | 47.83150 | 26.93861 | -0.06860921 |
| 12: | slant_down | 54.26785 | 16.76676 | 47.83590 | 26.93610 | -0.06897974 |
| 13: | wide_lines | 54.26692 | 16.77000 | 47.83160 | 26.93790 | -0.06657523 |

Datasaurus Dozen



Thank you!

# Functions of rv

Normal Distribution (Gaussian)  
Exponential distribution  
Uniform distribution

Check when to cover?